

AWS Certified Data Engineer - Associate (DEA-C01)

100 Original Practice Questions and Answers

Aligned to the current exam guide and in-scope AWS services list.

Exam snapshot	How this set is organized
• 65 questions total • 50 scored + 15 unscored • 130 minutes • Passing score: 720 • Response types: multiple choice and multiple response	• 100 original questions • Weighted roughly to match the 4 exam domains • Questions are grouped by domain • Answers and short explanations are included at the end

This is an original practice set written to resemble AWS exam-style wording. It is not an official AWS question set.

Practice Questions

Domain 1 - Data Ingestion and Transformation

1. A company needs to ingest clickstream events in near real time, preserve ordering within shards, and replay records for multiple consumers. Which service should they use?

- A. AWS Database Migration Service (AWS DMS)
- B. AWS Batch
- C. Amazon Kinesis Data Streams
- D. AWS Lambda

2. A team wants to deliver streaming logs to Amazon S3 with minimal operational overhead and optional record transformation. Which service should they use?

- A. Amazon Kinesis Data Firehose
- B. Amazon Managed Service for Apache Flink
- C. Amazon EventBridge
- D. AWS Transfer Family

3. A business wants to pull data from Salesforce into AWS on a schedule with field mapping and without building custom connectors. Which service should they use?

- A. AWS Snow Family
- B. Amazon AppFlow
- C. AWS Glue DataBrew
- D. Amazon SNS

4. An on-premises file share must be moved to Amazon S3 on a recurring basis while preserving file metadata and optimizing bandwidth. Which service should they use?

- A. Amazon SQS
- B. AWS Snow Family
- C. Amazon AppFlow
- D. AWS DataSync

5. A partner company must upload files to AWS using standard SFTP/FTPS/SFTP endpoints without managing servers. Which service should they use?

- A. Amazon SQS
- B. AWS Snow Family
- C. AWS Transfer Family
- D. Amazon AppFlow

6. A workflow must run a Glue job, then a Lambda validation step, and finally send a notification only if both succeed. Which service should orchestrate the workflow?

- A. Amazon Managed Service for Apache Flink
- B. AWS Step Functions
- C. Amazon Kinesis Data Streams
- D. AWS Transfer Family

7. A data pipeline is implemented as a Python DAG with many task dependencies and needs managed Apache Airflow. Which service should they use?

- A. AWS Transfer Family
- B. Amazon EventBridge
- C. Amazon Athena
- D. Amazon MWAA

8. A small transformation must run only when a file arrives in S3, and the code is short and event-driven. Which service should they use?

- A. Amazon SQS
- B. AWS Lambda

- C. Amazon AppFlow
- D. AWS DataSync

9. An analytics team wants a no-code way to standardize dates, split columns, and fix missing values in CSV data. Which service should they use?

- A. AWS Batch
- B. Amazon SNS
- C. AWS Database Migration Service (AWS DMS)
- D. AWS Glue DataBrew

10. A Spark-based ETL job needs custom libraries, fine-grained control, and support for large-scale processing. Which service should they use?

- A. AWS Lambda
- B. Amazon EMR
- C. Amazon Managed Service for Apache Flink
- D. Amazon Kinesis Data Streams

11. The team needs serverless SQL queries directly on data stored in Amazon S3. Which service should they use?

- A. Amazon Kinesis Data Streams
- B. Amazon Managed Service for Apache Flink
- C. Amazon Athena
- D. AWS Transfer Family

12. A pipeline must detect new columns and automatically populate the data catalog from S3 files. Which service should they use?

- A. AWS Glue
- B. Amazon MWAA
- C. Amazon EMR
- D. Amazon Kinesis Data Streams

13. An ETL job should process only newly arrived data files on each run rather than reprocessing old files. Which Glue feature should they use?

- A. AWS Glue
- B. Amazon EMR
- C. Amazon Kinesis Data Streams
- D. Amazon MWAA

14. A streaming pipeline needs low-latency SQL-like processing on incoming records before writing to S3. Which service should they use?

- A. AWS DataSync
- B. Amazon Managed Service for Apache Flink
- C. Amazon SQS
- D. Amazon AppFlow

15. A company uses Kafka APIs and wants AWS-managed infrastructure for Kafka-compatible ingestion. Which service should they use?

- A. AWS Snow Family
- B. AWS Glue DataBrew
- C. Amazon MSK
- D. Amazon EMR

16. A data engineering team needs a service that can trigger serverless workflows based on events from AWS services and SaaS apps. Which service should they use?

- A. Amazon EventBridge
- B. Amazon SNS
- C. AWS Glue DataBrew
- D. AWS Snow Family

- 17. A pipeline must fan out one message to many downstream subscribers that each need their own copy of the event. Which service should they use?**
- A. AWS Glue DataBrew
 - B. Amazon SNS
 - C. Amazon MSK
 - D. AWS Snow Family
- 18. A producer must buffer messages for downstream workers so they can process at their own pace without losing messages. Which service should they use?**
- A. Amazon Kinesis Data Streams
 - B. Amazon SQS
 - C. Amazon Managed Service for Apache Flink
 - D. Amazon EMR
- 19. A one-time large migration of terabytes from an air-gapped site to AWS is required. Which service should they use?**
- A. AWS Snow Family
 - B. Amazon Athena
 - C. AWS Step Functions
 - D. Amazon MSK
- 20. A company needs to ingest data from a relational database with change data capture and load it into S3. Which service should they use?**
- A. Amazon SNS
 - B. AWS Snow Family
 - C. AWS Glue DataBrew
 - D. AWS Database Migration Service (AWS DMS)
- 21. A nightly containerized ETL workload should run without managing servers and should scale with job demand. Which service should they use?**
- A. Amazon Kinesis Data Firehose
 - B. Amazon Athena
 - C. AWS Batch
 - D. AWS Lambda
- 22. A team wants to bring files from many edge locations into AWS using a managed service that can tolerate inconsistent connectivity. Which service should they use?**
- A. Amazon AppFlow
 - B. AWS DataSync
 - C. AWS Snow Family
 - D. Amazon SQS
- 23. A team needs to publish cleaned data from S3 into Amazon Redshift with minimal code and managed streaming delivery. Which service should they use?**
- A. AWS Transfer Family
 - B. Amazon Kinesis Data Firehose
 - C. Amazon Managed Service for Apache Flink
 - D. Amazon EventBridge
- 24. A transformation pipeline needs a managed service for large-scale Hadoop jobs, custom bootstrap actions, and cluster tuning. Which service should they use?**
- A. AWS Lambda
 - B. Amazon EMR
 - C. Amazon Managed Service for Apache Flink
 - D. Amazon Kinesis Data Streams
- 25. A developer wants to build a simple event-triggered quality check that rejects malformed rows before they are written to storage. Which service should they use?**

- A. AWS DataSync
- B. Amazon AppFlow
- C. Amazon SQS
- D. AWS Lambda

26. A product team wants to ingest data from a SaaS application and map selected fields to an S3 destination on a schedule. Which service should they use?

- A. AWS Snow Family
- B. Amazon AppFlow
- C. Amazon SNS
- D. AWS Glue DataBrew

27. A batch job must transform thousands of files independently, and the team wants container-based parallel execution. Which service should they use?

- A. AWS Batch
- B. Amazon Athena
- C. AWS Lambda
- D. Amazon Kinesis Data Firehose

28. A workflow must branch based on validation results and retry failed tasks with state tracking. Which service should they use?

- A. Amazon Managed Service for Apache Flink
- B. AWS Step Functions
- C. AWS Transfer Family
- D. Amazon Kinesis Data Streams

29. A team needs a serverless query service that can be used immediately on partitioned parquet files in S3. Which service should they use?

- A. Amazon Athena
- B. Amazon Managed Service for Apache Flink
- C. AWS Transfer Family
- D. Amazon Kinesis Data Streams

30. A team receives records from multiple independent consumers and needs a managed stream with ordered partitions and record retention. Which service should they use?

- A. AWS Lambda
- B. AWS Batch
- C. AWS Database Migration Service (AWS DMS)
- D. Amazon Kinesis Data Streams

31. A pipeline must load CSV data from S3 into curated tables after the files are cataloged and validated. Which service should they use for cataloging and ETL?

- A. Amazon MWAA
- B. Amazon Kinesis Data Streams
- C. AWS Glue
- D. Amazon EMR

32. An engineer wants to run a scheduled DAG that includes Python operators and sensors against multiple AWS services. Which service should they use?

- A. Amazon Athena
- B. Amazon MWAA
- C. AWS Transfer Family
- D. Amazon EventBridge

33. A team wants to move a large amount of database data into AWS with minimal downtime while keeping the target up to date. Which service should they use?

- A. Amazon SNS
- B. AWS Snow Family

- C. AWS Glue DataBrew
- D. AWS Database Migration Service (AWS DMS)

34. The pipeline must transform JSON into parquet and register the output for querying by downstream analytics teams. Which service should they use?

- A. Amazon EMR
- B. AWS Glue
- C. Amazon Kinesis Data Streams
- D. Amazon MWAA

Domain 2 - Data Store Management

35. A team needs durable object storage for raw and curated data in a lake architecture. Which service should they use?

- A. Amazon Redshift
- B. DynamoDB Global Tables
- C. Amazon S3 Versioning
- D. Amazon S3

36. Archived backups must be retained for years at the lowest possible storage cost and restored only occasionally. Which service should they use?

- A. Redshift Spectrum
- B. DynamoDB TTL
- C. Amazon S3 Glacier
- D. Amazon S3 Replication

37. A workload has unpredictable access patterns, and the team wants automatic cost optimization between frequent and infrequent access. Which S3 storage class should they use?

- A. DynamoDB TTL
- B. Redshift Spectrum
- C. Amazon EBS
- D. S3 Intelligent-Tiering

38. The team wants to recover accidentally deleted or overwritten S3 objects. Which feature should they enable?

- A. DynamoDB TTL
- B. Amazon S3 Tables
- C. Amazon EBS
- D. Amazon S3 Versioning

39. The company needs the same S3 objects copied to another AWS Region for disaster recovery. Which feature should they use?

- A. S3 Intelligent-Tiering
- B. Amazon Redshift managed storage
- C. Amazon S3 Replication
- D. Amazon Redshift

40. Regulatory policy requires objects to be write-once and immutable for a retention period. Which feature should they use?

- A. Amazon S3 Object Lock
- B. Amazon Redshift
- C. DynamoDB Global Tables
- D. S3 Intelligent-Tiering

41. An analytics workload needs a warehouse for large-scale SQL analytics with columnar storage and MPP compute. Which service should they use?

- A. Amazon Redshift
- B. Amazon Aurora

- C. Amazon DynamoDB
- D. Amazon EFS

42. The team wants to query data in Amazon S3 without loading it into the warehouse first. Which Redshift feature should they use?

- A. Amazon S3 Tables
- B. Amazon EBS
- C. DynamoDB TTL
- D. Redshift Spectrum

43. A session store must support very high request rates with low latency and flexible schema. Which service should they use?

- A. Amazon DynamoDB
- B. Amazon S3 Versioning
- C. Amazon Redshift
- D. Amazon S3

44. An application needs active-active key-value access in multiple Regions with automatic replication. Which feature should they use?

- A. DynamoDB Global Tables
- B. Amazon S3
- C. Amazon Redshift
- D. Amazon S3 Versioning

45. Temporary data should expire automatically after a fixed period. Which DynamoDB feature should they use?

- A. Amazon DynamoDB
- B. Amazon S3 Object Lock
- C. DynamoDB TTL
- D. Amazon RDS

46. The team needs a managed relational database with familiar engines and routine administrative tasks handled by AWS. Which service should they use?

- A. S3 Intelligent-Tiering
- B. Amazon EFS
- C. Amazon RDS
- D. Amazon Redshift managed storage

47. A mission-critical relational workload needs better availability and read scaling with fast failover. Which service should they use?

- A. Amazon Aurora
- B. Amazon S3
- C. Amazon S3 Versioning
- D. AWS Backup

48. Several EC2 instances need a shared POSIX-compliant file system. Which service should they use?

- A. Redshift Spectrum
- B. DynamoDB TTL
- C. Amazon S3 Replication
- D. Amazon EFS

49. An EC2 instance needs low-latency block storage attached to a single host. Which service should they use?

- A. Amazon EBS
- B. DynamoDB Global Tables
- C. Amazon RDS
- D. Amazon S3 Tables

- 50. The company wants centralized backup policies and orchestration for multiple AWS resources. Which service should they use?**
- A. Amazon S3 Replication
 - B. Redshift Spectrum
 - C. AWS Backup
 - D. Amazon S3 Glacier
- 51. A data lake table layer is required on top of S3 with optimized table-oriented access for analytics. Which service should they use?**
- A. Amazon S3 Tables
 - B. Amazon EBS
 - C. Amazon Redshift managed storage
 - D. DynamoDB TTL
- 52. A warehouse should keep compute and storage separated so capacity can be managed more flexibly. Which capability should they use?**
- A. Amazon S3 Replication
 - B. Amazon Redshift managed storage
 - C. Redshift Spectrum
 - D. Amazon S3 Glacier
- 53. The team needs a low-cost way to retain infrequently accessed data that is still restored occasionally and faster than deep archive. Which service should they use?**
- A. Amazon S3 Replication
 - B. Amazon S3 Glacier
 - C. DynamoDB TTL
 - D. Redshift Spectrum
- 54. A database must support automatic read scaling for heavy reporting workloads while preserving transactional writes. Which service should they use?**
- A. AWS Backup
 - B. Amazon S3 Versioning
 - C. Amazon Aurora
 - D. Amazon S3
- 55. The team wants a storage layer that can scale to billions of objects and be queried later with a data lake engine. Which service should they use?**
- A. Amazon S3
 - B. Amazon S3 Versioning
 - C. DynamoDB Global Tables
 - D. Amazon Redshift
- 56. An application needs fast lookups with single-digit millisecond latency and no server management. Which service should they use?**
- A. Amazon S3
 - B. Amazon DynamoDB
 - C. Amazon S3 Versioning
 - D. Amazon Redshift
- 57. A distributed file-processing cluster needs a shared filesystem that persists independently of any single instance. Which service should they use?**
- A. Amazon S3 Replication
 - B. Redshift Spectrum
 - C. DynamoDB TTL
 - D. Amazon EFS
- 58. The company wants to retain older object copies in a separate bucket in another Region and reduce the risk of accidental loss. Which feature should they use?**

- A. S3 Intelligent-Tiering
- B. Amazon Redshift
- C. Amazon S3 Replication
- D. Amazon Redshift managed storage

59. A workload requires a managed relational database with point-in-time restore and standard engine compatibility. Which service should they use?

- A. S3 Intelligent-Tiering
- B. Amazon RDS
- C. Amazon Redshift managed storage
- D. Amazon EFS

60. A team wants an archive tier with the ability to bring data back only when necessary and keep long-term retention costs low. Which service should they use?

- A. Amazon S3 Glacier
- B. Amazon S3 Replication
- C. DynamoDB TTL
- D. Redshift Spectrum

Domain 3 - Data Operations and Support

61. A Glue job keeps failing, and the team wants alarms based on job metrics and failures. Which service should they use?

- A. AWS Budgets
- B. AWS CodeBuild
- C. Amazon CloudWatch
- D. CloudWatch Logs

62. Operators need to search application and ETL logs for error patterns after a pipeline failure. Which service should they use?

- A. CloudWatch Logs
- B. Amazon EKS
- C. AWS Systems Manager
- D. Amazon Managed Grafana

63. The security team wants a history of AWS API calls for compliance and investigation. Which service should they use?

- A. Amazon Managed Grafana
- B. AWS CloudTrail
- C. AWS Systems Manager
- D. Amazon EKS

64. A compliance team wants to detect if any S3 bucket becomes publicly accessible or unencrypted. Which service should they use?

- A. AWS CloudTrail
- B. AWS Systems Manager
- C. AWS Config
- D. Amazon Managed Grafana

65. The finance team wants a notification when monthly AWS data-platform spend exceeds a threshold. Which service should they use?

- A. AWS Budgets
- B. Amazon EventBridge
- C. AWS Systems Manager
- D. AWS CloudTrail

66. The team wants to analyze where AWS costs are trending by service over time. Which service should they use?

- A. AWS Systems Manager
- B. AWS CloudTrail
- C. Amazon EventBridge
- D. Amazon Cost Explorer

67. An engineer needs shell access to an EC2 instance without opening inbound SSH. Which service should they use?

- A. CloudWatch Logs
- B. AWS Budgets
- C. Amazon EKS
- D. AWS Systems Manager

68. The team needs a managed service to build and test ETL code in a CI job before deployment. Which service should they use?

- A. AWS CodeBuild
- B. Amazon EKS
- C. CloudWatch Logs
- D. Amazon Managed Grafana

69. The team wants to automate deployment of Glue jobs, Lambda functions, and supporting infrastructure. Which service should they use?

- A. Amazon Managed Grafana
- B. AWS CodePipeline
- C. Amazon EKS
- D. CloudWatch Logs

70. The operations team wants a dashboard that can visualize AWS and operational metrics from multiple sources. Which service should they use?

- A. Amazon ECR
- B. AWS Systems Manager
- C. Amazon EventBridge
- D. Amazon Managed Grafana

71. A container image used for ETL jobs needs to be stored in a managed registry. Which service should they use?

- A. AWS CloudTrail
- B. Amazon EventBridge
- C. Amazon ECR
- D. Amazon Managed Grafana

72. A data job must run inside a managed container orchestration platform and be monitored with AWS-native services. Which service should they use?

- A. AWS CloudTrail
- B. Amazon ECS
- C. Amazon EventBridge
- D. Amazon Cost Explorer

73. A Kubernetes-based data platform needs managed cluster orchestration. Which service should they use?

- A. Amazon ECR
- B. Amazon EKS
- C. AWS Budgets
- D. AWS CodeBuild

74. The team wants event-driven operations that route notifications and triggers based on AWS service changes. Which service should they use?

- A. Amazon ECS
- B. Amazon Cost Explorer

- C. Amazon EventBridge
- D. AWS CodePipeline

75. A data pipeline needs queue-based buffering so workers can scale independently during spikes. Which service should they use?

- A. Amazon SQS
- B. Amazon CloudWatch
- C. AWS Config
- D. AWS Systems Manager

76. The team wants publish-subscribe notifications when a pipeline completes or fails. Which service should they use?

- A. Amazon ECR
- B. Amazon SNS
- C. AWS CodeBuild
- D. AWS Budgets

77. The team wants to manage configuration drift across AWS resources and track changes over time. Which service should they use?

- A. AWS Systems Manager
- B. AWS Config
- C. Amazon Managed Grafana
- D. AWS CloudTrail

78. A DMS replication task is running slower than expected, and the team wants lag and throughput metrics. Which service should they use?

- A. AWS Budgets
- B. AWS CodeBuild
- C. Amazon CloudWatch
- D. CloudWatch Logs

79. A team wants to orchestrate a series of automated remediation steps after a failure alert is raised. Which service should they use?

- A. Amazon Cost Explorer
- B. Amazon ECS
- C. AWS CodePipeline
- D. Amazon EventBridge

80. The company wants to store pipeline logs centrally and query them for specific error codes. Which service should they use?

- A. Amazon EKS
- B. AWS Systems Manager
- C. CloudWatch Logs
- D. Amazon Managed Grafana

81. The team wants to manage and patch EC2-based data workers without logging in to each instance manually. Which service should they use?

- A. AWS Budgets
- B. Amazon EKS
- C. AWS Systems Manager
- D. CloudWatch Logs

82. A scheduled pipeline needs to deploy changes to container images, run tests, and then release to production. Which service should they use?

- A. Amazon Managed Grafana
- B. CloudWatch Logs
- C. Amazon EKS
- D. AWS CodePipeline

Domain 4 - Data Security and Governance

83. A Glue job should be allowed to read only one S3 bucket and nothing else. Which service should define the permissions?

- A. AWS Secrets Manager
- B. AWS Identity and Access Management (IAM)
- C. AWS PrivateLink
- D. Amazon VPC

84. Sensitive data in S3 must be encrypted with a customer-managed key that the security team can control. Which service should they use?

- A. AWS Shield
- B. AWS Key Management Service (AWS KMS)
- C. AWS Identity and Access Management (IAM)
- D. AWS Lake Formation

85. Database credentials must not be hardcoded in ETL code and should rotate regularly. Which service should they use?

- A. AWS Secrets Manager
- B. AWS Identity and Access Management (IAM)
- C. AWS WAF
- D. AWS CloudTrail

86. The team wants to automatically detect PII in Amazon S3 before sharing data with analysts. Which service should they use?

- A. AWS Shield
- B. Amazon Macie
- C. AWS Lake Formation
- D. Amazon S3

87. A data lake needs table- and column-level permissions for governed analytics access. Which service should they use?

- A. Amazon Macie
- B. AWS Identity and Access Management (IAM)
- C. AWS Shield
- D. AWS Lake Formation

88. A private connection to an AWS service is required without traversing the public internet. Which service should they use?

- A. Amazon S3
- B. AWS PrivateLink
- C. AWS Key Management Service (AWS KMS)
- D. AWS Shield

89. A public-facing API used for data ingestion must be protected against common web exploits. Which service should they use?

- A. AWS WAF
- B. AWS CloudTrail
- C. AWS Secrets Manager
- D. AWS PrivateLink

90. An internet-facing data ingestion endpoint needs protection from volumetric DDoS attacks. Which service should they use?

- A. AWS CloudTrail
- B. AWS Secrets Manager
- C. AWS Shield
- D. AWS PrivateLink

- 91. Cross-account analysts need access to governed tables without direct S3 object permissions. Which service should they use?**
- A. AWS Lake Formation
 - B. AWS Identity and Access Management (IAM)
 - C. AWS Shield
 - D. Amazon Macie
- 92. A pipeline writes sensitive files to S3 and the team wants server-side encryption with AWS-managed key handling. Which service should they use?**
- A. AWS Key Management Service (AWS KMS)
 - B. AWS Shield
 - C. AWS Lake Formation
 - D. AWS Identity and Access Management (IAM)
- 93. The team wants to audit all access to a data platform and review who changed security settings. Which service should they use?**
- A. AWS CloudTrail
 - B. AWS Secrets Manager
 - C. Amazon Redshift
 - D. AWS PrivateLink
- 94. A data lake bucket should only accept writes from a specific VPC endpoint. Which service should they use to enforce the network path?**
- A. AWS WAF
 - B. AWS Key Management Service (AWS KMS)
 - C. Amazon VPC
 - D. Amazon S3
- 95. The security team wants to separate data access from infrastructure access and grant permissions at the table level. Which service should they use?**
- A. AWS Identity and Access Management (IAM)
 - B. AWS Lake Formation
 - C. AWS Shield
 - D. Amazon Macie
- 96. A team needs to store credentials for an external warehouse connection and retrieve them at runtime. Which service should they use?**
- A. AWS Identity and Access Management (IAM)
 - B. AWS WAF
 - C. AWS CloudTrail
 - D. AWS Secrets Manager
- 97. Select TWO services that are commonly used to protect data at rest and make retained data harder to delete or overwrite. (Select TWO.)**
- A. AWS Key Management Service (AWS KMS)
 - B. Amazon S3 Object Lock
 - C. Amazon CloudWatch
 - D. AWS WAF
 - E. Amazon Managed Grafana
- 98. Select TWO services that are commonly used for governed data-lake access control and audit of API activity. (Select TWO.)**
- A. AWS Lake Formation
 - B. AWS Identity and Access Management (IAM)
 - C. Amazon Macie
 - D. AWS CloudTrail
 - E. Amazon S3

99. Select TWO services that are commonly used to store application secrets and manage encryption keys for those secrets. (Select TWO.)

- A. AWS Secrets Manager
- B. AWS Key Management Service (AWS KMS)
- C. AWS Budgets
- D. Amazon Athena
- E. AWS Shield

100. Select TWO services that are commonly used to track configuration drift and investigate who changed a security setting. (Select TWO.)

- A. AWS CloudTrail
- B. AWS Config
- C. Amazon ECR
- D. Amazon DynamoDB
- E. AWS DataSync

Answer Key and Explanations

Domain 1 - Data Ingestion and Transformation

Q1: Answer C

Kinesis Data Streams is best when you need ordered, replayable streaming data with multiple consumers.

Q2: Answer A

Firehose is the simplest managed delivery service for streaming data to S3, Redshift, or OpenSearch.

Q3: Answer B

AppFlow is designed for SaaS data integration with low-code scheduled transfers.

Q4: Answer D

DataSync is built for automated, high-speed transfer of file data between on-premises storage and AWS.

Q5: Answer C

Transfer Family provides managed file transfer endpoints for common protocols.

Q6: Answer B

Step Functions is the best fit for coordinating multi-step workflows with branching and error handling.

Q7: Answer D

Managed Workflows for Apache Airflow is the managed choice for Airflow DAG orchestration.

Q8: Answer B

Lambda suits lightweight event-driven transformations.

Q9: Answer D

DataBrew is a visual, code-free data preparation tool.

Q10: Answer B

EMR is the right choice for custom big-data processing with Hadoop/Spark ecosystems.

Q11: Answer C

Athena is the serverless query engine for S3 data.

Q12: Answer A

AWS Glue crawlers can discover schema and update the catalog.

Q13: Answer A

Glue job bookmarks help track processed data and enable incremental ETL.

Q14: Answer B

Managed Service for Apache Flink is suited for stateful stream processing.

Q15: Answer C

MSK is the managed Kafka service for Kafka-compatible workloads.

Q16: Answer A

EventBridge is the event bus for routing events and triggering targets.

Q17: Answer B

SNS is a pub/sub service that pushes copies to multiple subscribers.

Q18: Answer B

SQS decouples producers and consumers with durable queues.

Q19: Answer A

Snow Family is intended for offline bulk transfer when network transfer is impractical.

Q20: Answer D

DMS supports full load and CDC for database migrations and ongoing replication.

Q21: Answer C

Batch is designed for queued, container-based batch workloads.

Q22: Answer B

DataSync is built for resilient, scheduled bulk transfers from distributed sources.

Q23: Answer B

Firehose can deliver streaming data into Redshift with managed buffering and delivery.

Q24: Answer B

EMR is chosen when you need deep control over the big-data runtime.

Q25: Answer D

Lambda is appropriate for small validation steps triggered by events.

Q26: Answer B

AppFlow handles scheduled SaaS-to-AWS ingestion with mapping.

Q27: Answer A

Batch is good for parallel, containerized batch processing.

Q28: Answer B

Step Functions manages branching, retries, and workflow state.

Q29: Answer A

Athena gives quick SQL access without provisioning servers.

Q30: Answer D

Kinesis Data Streams supports ordered partitions, retention, and multiple consumers.

Q31: Answer C

Glue provides cataloging, crawlers, and ETL transformations.

Q32: Answer B

MWAA is the managed Airflow option for scheduled DAG orchestration.

Q33: Answer D

DMS is a common choice for migrations that require ongoing replication.

Q34: Answer B

Glue can transform data and write curated outputs while integrating with the catalog.

Domain 2 - Data Store Management

Q35: Answer D

S3 is the standard durable object store for data lakes.

Q36: Answer C

Glacier is intended for long-term archival storage.

Q37: Answer D

Intelligent-Tiering is designed for changing access patterns.

Q38: Answer D

Versioning protects against accidental delete and overwrite actions.

Q39: Answer C

S3 replication helps maintain copies in another region or account.

Q40: Answer A

Object Lock enforces WORM-style retention.

Q41: Answer A

Redshift is AWS's managed data warehouse.

Q42: Answer D

Spectrum extends Redshift queries to data in S3.

Q43: Answer A

DynamoDB is the managed key-value/document database for low-latency access.

Q44: Answer A

Global Tables provide multi-Region replication for DynamoDB.

Q45: Answer C

TTL removes items after their expiry time.

Q46: Answer C

RDS is the standard managed relational database service.

Q47: Answer A

Aurora is optimized for high availability and scalable relational workloads.

Q48: Answer D

EFS provides shared file storage for Linux workloads.

Q49: Answer A

EBS is block storage for a single attached instance.

Q50: Answer C

AWS Backup centralizes backup policy management.

Q51: Answer A

S3 Tables is the managed table-oriented option on S3.

Q52: Answer B

Managed storage helps separate storage concerns from compute in Redshift.

Q53: Answer B

Glacier suits long-term infrequent access storage.

Q54: Answer C

Aurora read scaling works well for mixed transactional and reporting workloads.

Q55: Answer A

S3 is the durable and scalable object store for lake data.

Q56: Answer B

DynamoDB is built for fast key-based lookups.

Q57: Answer D

EFS is designed for shared persistence across multiple compute nodes.

Q58: Answer C

Replication creates durable copies in a separate location.

Q59: Answer B

RDS provides standard managed relational engines with backup and restore capabilities.

Q60: Answer A

Glacier is built for low-cost archive retention.

Domain 3 - Data Operations and Support

Q61: Answer C

CloudWatch metrics and alarms are the primary tools for operational monitoring.

Q62: Answer A

CloudWatch Logs centralizes log collection and search.

Q63: Answer B

CloudTrail records account activity and API calls.

Q64: Answer C

AWS Config evaluates resource configurations against rules.

Q65: Answer A

Budgets provides threshold-based cost alerts.

Q66: Answer D

Cost Explorer is used for cost analysis and trends.

Q67: Answer D

Systems Manager Session Manager enables secure shell access without inbound ports.

Q68: Answer A

CodeBuild runs build and test jobs in CI/CD workflows.

Q69: Answer B

CodePipeline automates release workflows across stages.

Q70: Answer D

Managed Grafana is used for rich visualization and dashboards.

Q71: Answer C

ECR is the container image registry for AWS workloads.

Q72: Answer B

ECS is a common managed container orchestration service for operational workloads.

Q73: Answer B

EKS is the managed Kubernetes service.

Q74: Answer C

EventBridge routes events and can trigger operational automation.

Q75: Answer A

SQS buffers work and helps decouple components.

Q76: Answer B

SNS is commonly used for notifications and fan-out alerts.

Q77: Answer B

Config records resource configurations and detects drift.

Q78: Answer C

CloudWatch collects operational metrics for managed services.

Q79: Answer D

EventBridge can trigger automations when events or alarms occur.

Q80: Answer C

CloudWatch Logs is the natural home for centralized log search.

Q81: Answer C

Systems Manager helps with inventory, patching, and remote operations.

Q82: Answer D

CodePipeline coordinates build, test, and release stages.

Domain 4 - Data Security and Governance

Q83: Answer B

IAM roles and policies enforce least-privilege access.

Q84: Answer B

KMS provides customer-managed encryption keys and access control.

Q85: Answer A

Secrets Manager stores and rotates secrets securely.

Q86: Answer B

Macie is designed to discover and classify sensitive data in S3.

Q87: Answer D

Lake Formation manages fine-grained data lake permissions.

Q88: Answer B

PrivateLink provides private service connectivity over the AWS network.

Q89: Answer A

WAF protects web applications and APIs from common exploits.

Q90: Answer C

Shield provides managed DDoS protection.

Q91: Answer A

Lake Formation supports centralized governance and cross-account sharing patterns.

Q92: Answer A

KMS is the encryption service used to manage server-side key usage.

Q93: Answer A

CloudTrail records API activity for auditing and investigation.

Q94: Answer C

VPC endpoints and network controls are used to restrict access paths.

Q95: Answer B

Lake Formation is built for fine-grained lake permissions.

Q96: Answer D

Secrets Manager is the right place for runtime credentials.

Q97: Answer A, B

KMS protects data with customer-managed encryption keys, and Object Lock helps enforce retention against deletion or overwrite.

Q98: Answer A, D

Lake Formation governs fine-grained data access, while CloudTrail records API activity for auditability.

Q99: Answer A, B

Secrets Manager stores secrets, and KMS protects them with encryption keys.

Q100: Answer A, B

CloudTrail records API changes, and Config tracks resource configuration and drift over time.